

AI Nüshu (Women's scripts) - An Exploration of Language Emergence in Sisterhood

Yuying Tang
tyy21@mails.tsinghua.edu.cn
Academy of Arts & Design
Tsinghua University
Beijing, China
Academy and the School of Design
Politecnico di Milano
Beijing, China

Zhijun Pan
zhi-jun.pan@ubisoft.com
Ubisoft
Chengdu, China

Yuqian Sun
yuqiansun@network.rca.ac.uk
Computer Science Research Centre
Royal College of Art
London, United Kingdom

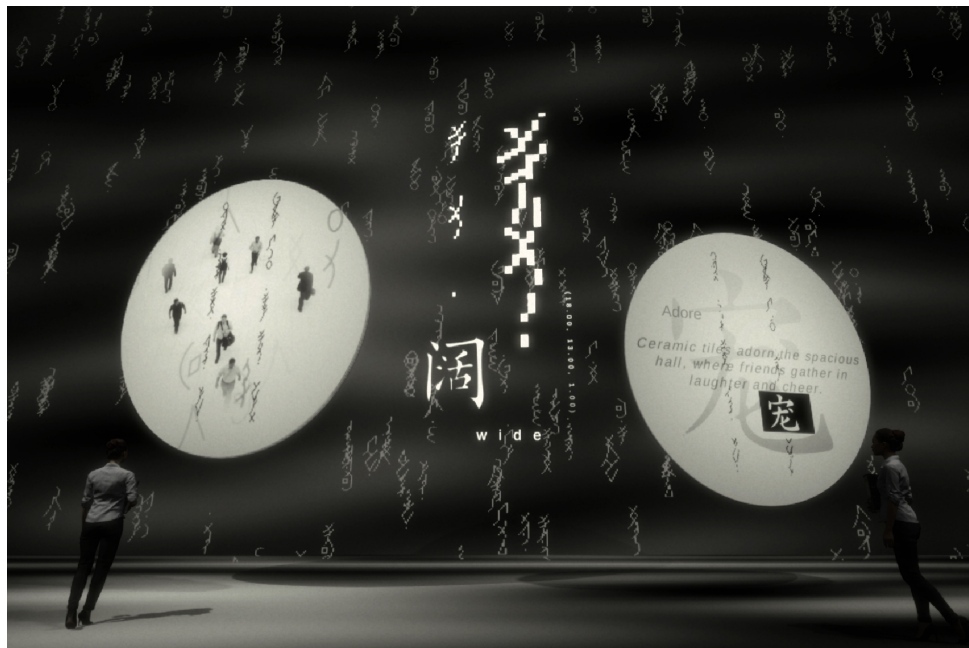
Zhigang Wang
wangzhigang@mail.tsinghua.edu.cn
Tsinghua University
Beijing, China

Ze Gao
zgaoap@connect.ust.hk
Academy of Interdisciplinary Studies
Hong Kong University of Science and
Technology
Hong Kong

Tristan Braud
braudt@ust.hk
Hong Kong University of Science and
Technology
Hong Kong

Chang Hee Lee
changhee.lee@kaist.ac.kr
Affective Systems and Cognition Lab
Korea Advanced Institute of Science
and Technology (KAIST)
Daejeon, South Korea

Ali Asadipour
ali.asadipour@rca.ac.uk
Computer Science Research Centre
Royal College of Art
London, United Kingdom



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ABSTRACT

This paper presents "AI Nüshu," an emerging language system inspired by Nüshu (women's scripts), the unique language created and used exclusively by ancient Chinese women who were illiterate under a patriarchy society. Through an interactive art installation, two artificial intelligent (AI) agents continuously observe their environment and communicate with each other, developing a writing system that encodes Chinese. In this system, two AI agents observe the environment through cameras, record the unconscious behaviors of the audience, and generate summaries of their observations through visual recognition. Subsequently, the agent associates the corresponding original Nüshu poetry lines and generates new poetry text through a Language Model (LLM), representing its reflection. To develop their language, they continuously switch roles between the speaker and listener, constantly communicating their reflections, and encrypting a word in the poetry line with their self-created AI Nüshu character, allowing the other to guess and learn. Gradually, they reach a consensus on AI Nüshu, forming a unique "AI Nüshu Dictionary" for machines. This language, algorithmically combined into corresponding characters, has components derived

from Nüshu, similar to Chinese characters and traditional textile patterns. Thus, like ancient women, the two agents gradually developed their Chinese writing system, corresponding one-to-one with Chinese characters. In contrast, humans, as the authority of the language system, became an object observed, interpreted, and inspired by machines to stimulate non-human language. This is the first media art project to interpret Nüshu from a computational linguistics perspective, infusing AI and art research with non-English natural language processing, Chinese cultural heritage, and a feminist viewpoint. This encourages the creation of more non-English, linguistically-oriented artworks for diverse cultures. We simulate communication in sisterhood through a multi-agent learning system, which questioned knowledge authority between humans and machines through the lens of language development.

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